NETWORKING PRACTICAL ASSIGNMENT

Windows Command Prompt • Ping • Network Monitoring • Wireshark • AI Troubleshooting

Student Name: _____ **Class / Batch:** _____

Date: _____ **Subject:** Networking

Important: Perform the practical tasks on your own computer. Add your own screenshots and write the actual values you see. The blank spaces below are for your results; do not copy made-up IP addresses or bandwidth readings.

1. Find your network configuration using ipconfig /all

Steps

1. Connect your computer to Wi-Fi or Ethernet.
2. Press **Windows + R**, type **cmd**, and press Enter.
3. Type **ipconfig /all** and press Enter.
4. Find the active Wi-Fi or Ethernet adapter. Note the IPv4 Address, Subnet Mask, Default Gateway, and DNS Servers.
5. Take a screenshot of the Command Prompt output and paste it below or attach it to this assignment.

Network detail	Your computer's result
IPv4 Address	_____
Subnet Mask	_____
Default Gateway	_____
DNS Server(s)	_____

Screenshot: Paste or attach your ipconfig /all screenshot here.

_____

_____

2. Test connectivity using ping

1. Open Command Prompt.
2. Disconnect Wi-Fi or unplug the Ethernet cable. Then run: **ping www.google.com**
3. Note the message or result. Reconnect to the internet and run the same command again.
4. Compare the two results. Take screenshots if your teacher requires them.

What the results mean

• **Reply from …** — A response was received; the destination is reachable. The time shows the round-trip delay.
• **Request timed out** — No reply arrived before the time limit. This can happen because of packet loss, filtering, or an unavailable destination.
• **Destination host unreachable** — Your computer or a network device reports that it cannot reach the destination.

• **Could not find host** — The name could not be resolved, often because DNS or the connection is unavailable.

Test	What appeared on your screen?
Wi-Fi / Ethernet disconnected	_____ _____
After reconnecting	_____ _____

Short note / difference: _____
_____

3. Monitor network usage while streaming YouTube

1. Connect to the internet and open a YouTube video.

2. Press **Ctrl + Shift + Esc** to open Task Manager. Select **Performance** → **Wi-Fi** (or Ethernet) to view network activity. You can also use the Processes tab and check the Network column.

3. Play the video for 5 minutes. Watch the network graph and note the highest value shown. Make sure you record the unit (for example, Mbps or Kbps).

4. In the Processes tab, identify the application/process showing the highest network use. The process name may vary by browser and system.

Observation	Your result
Peak network usage (include unit)	_____
Process using the most bandwidth	_____
Video / browser used	_____

Screenshot: Attach a Task Manager / Resource Monitor screenshot showing network activity.

4. Capture and filter traffic using Wireshark

Wireshark is a packet-analysis tool. Use it only on your own device/network or a network you have permission to inspect.

1. Download Wireshark from its official website: **https://www.wireshark.org/**. Install it if permitted. During installation, allow the packet-capture component (Npcap) if prompted.

2. Open Wireshark and choose the active Wi-Fi or Ethernet interface. Start capturing.

3. Browse Flipkart or Instagram Web for at least 30 seconds, then stop the capture.

4. In the display-filter bar, enter **http** to show HTTP packets. For HTTPS/TLS traffic, use **tls** (or **tcp.port == 443**) because modern HTTPS traffic is encrypted and the display filter keyword **https** is generally not the correct Wireshark filter.

5. Select a packet and inspect the packet details pane. Take a screenshot of the filtered packet list.

Three types of information visible in filtered packets

1. **Source and destination information:** source/destination IP addresses, and sometimes MAC addresses.

2. **Protocol and connection information:** protocol type, TCP ports, packet length, and timing.

3. **HTTP details when unencrypted HTTP is present:** request method (such as GET), host name, URL path, and response status. With HTTPS, the content is encrypted; you can usually see connection metadata such as IPs, ports, packet sizes, and TLS handshake details, not the private page contents.

My observation / packet details:

_____

_____

Screenshot: Attach your Wireshark capture and filter screenshot.

5. AI-generated checklist: Improve home network speed for online gaming

Scenario: A friend says online games are lagging or disconnecting. The following checklist is an AI-generated response.

• **1. Use Ethernet:** Connect the gaming PC or console to the router with an Ethernet cable when possible. It is usually more stable than Wi-Fi.

• **2. Improve Wi-Fi signal:** Keep the router in an open, central, elevated place, away from thick walls and interference.

• **3. Choose the right Wi-Fi band:** Use 5 GHz or 6 GHz when close to the router for lower interference; use 2.4 GHz when longer range is needed.

• **4. Reduce background traffic:** Pause large downloads, cloud backups, and 4K streaming while gaming.

• **5. Restart and update:** Restart the router occasionally and install router firmware and network-adapter updates from trusted sources.

• **6. Check latency and packet loss:** Run ping tests to the router and a reliable internet host. High ping or packet loss can indicate a Wi-Fi, router, or ISP problem.

• **7. Check router settings:** Enable QoS / device prioritization if supported, and secure the Wi-Fi so unknown devices cannot use the connection.

• **8. Contact the ISP if needed:** If lag continues on a wired connection, ask the ISP to check line quality, outages, and routing.

Suggestion I would try first: Use a wired Ethernet connection for gaming, because it can reduce wireless interference and make the connection more stable.

My chosen suggestion and reason:

_____

_____

Conclusion

In this practical assignment, I learned how to check network settings using ipconfig, test connectivity using ping, monitor network usage, capture and filter packets in Wireshark, and use an AI-generated checklist to troubleshoot gaming network problems.

Student Signature: _____ Date: _____